

Chapter 9 — Uniprocessor Scheduling

Model Answers for Exercises 1–10

FCFS, RR, SPN, SRT, and HRRN — Scheduling Diagrams and Normalized Turnaround Time Analysis

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Introduction

This document provides full model answers for the 10 short-term uniprocessor scheduling exercises in Chapter 9. For each example, five scheduling algorithms, First-Come-First-Served (FCFS), Round-Robin (RR) with the specified quantum, Shortest Process Next (SPN), Shortest Remaining Time (SRT), and Highest Response Ratio Next (HRRN), are applied. For each algorithm, the document presents a tabular Gantt chart (one cell per time unit), a per-process metrics table (arrival, service, finish, turnaround time TAT, and normalized turnaround time NTAT), and the average NTAT. A final critical analysis section compares the algorithms across all 10 examples and discusses the practical implications.

Background: Scheduling Algorithms and Performance Metric

This section establishes the notation and the five scheduling policies analyzed throughout the document. The terminology follows Stallings, Operating Systems: Internals and Design Principles, Chapter 9.

Performance Metric: Normalized Turnaround Time (NTAT)

For a process p with arrival time $A(p)$, service time $S(p)$, and finish time $F(p)$:

$$\text{Turnaround Time: } TAT(p) = F(p) - A(p)$$

$$\text{Normalized Turnaround Time: } N_{TAT}(P) = \frac{TAT(p)}{S(p)}$$

N_{TAT} is the ratio of total time spent in the system to the actual CPU time required. A value of 1.0 is the theoretical minimum (no waiting); higher values indicate proportionally more queuing delay. N_{TAT} is preferred over plain TAT for comparison, because it is dimensionless and treats short and long jobs on the same scale, in which a short job forced to wait a long time is penalized more, which reflects user-perceived responsiveness.

The average N_{TAT} across all processes is the headline number used to compare algorithms in this exercise set.

Algorithms

First-Come-First-Served (FCFS)

Non-preemptive. Processes are dispatched in arrival order; once selected a process runs to completion. Simple to implement but biased against short jobs that arrive behind a long one (the convoy effect).

Round-Robin (RR) with quantum q

Preemptive. The ready queue is FIFO; each dispatched process runs for at most q time units, then re-enters the tail of the queue if it has remaining work. Provides fairness and responsiveness independent of service time. Performance depends critically on q : too small $\rightarrow$ high context-switching overhead; too large $\rightarrow$ degenerates toward FCFS. In this exercise context-switch cost is assumed to be zero, so RR's only penalty is rotation delay.

Shortest Process Next (SPN)

Non-preemptive. At each dispatch decision, the ready process with the smallest service time is selected. Minimizes average waiting time over a fixed set of jobs known in advance, but risks starvation of long jobs and requires service-time estimates (assumed known here).

Shortest Remaining Time (SRT)

Preemptive variant of SPN. Whenever a new process arrives or the running process is interrupted, the scheduler picks the process with the smallest remaining service time. Optimal for average turnaround time when all service times are known, at the cost of even greater starvation risk for long jobs.

Highest Response Ratio Next (HRRN)

Non-preemptive. At each dispatch decision, the ready process with the highest response ratio $R = (w + s) / s$ is selected, where w is the time already spent waiting and s is the service time. Favors short jobs initially, but the ratio of long-waiting jobs grows with w , preventing starvation. Often described as a balanced compromise between SPN and FCFS.

Tie-breaking Rules Used

Following the assumptions in the exercise:

- (1) on equal arrival, FCFS uses alphabetical order;
- (2) for SPN and SRT ties, FCFS order (earlier arrival first) is applied, then alphabetical;
- (3) for HRRN ties (rare in integer schedules), earlier arrival wins.

Example 1

Round-Robin quantum: $q = 1$. Total service time = 15. Processes: 3.

Process Set

Process	Arrival	Service
A	0	5
B	2	3
C	5	7

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(3+3)/3 = 2.00$
C	$(7+3)/7 = 1.43$
Average	1.48

Round-Robin (RR, q = 1)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+4)/5 = 1.80$
B	$(3+3)/3 = 2.00$
C	$(7+3)/7 = 1.43$
Average	1.74

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(3+3)/3 = 2.00$
C	$(7+3)/7 = 1.43$
Average	1.48

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(3+3)/3 = 2.00$
C	$(7+3)/7 = 1.43$
Average	1.48

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(3+3)/3 = 2.00$
C	$(7+3)/7 = 1.43$
Average	1.48

Algorithm Comparison for Example 1

Algorithm	Average Normalized Turnaround Time
FCFS	1.48
RR	1.74
SPN	1.48
SRT	1.48
HRRN	1.48

Observations

The best algorithm for this workload is FCFS (avg NTAT = 1.48), while RR (avg NTAT = 1.74) is the worst.

RR ($q = 1$) performs worse than FCFS here because the quantum induces frequent context switching, which lengthens the turnaround time of short jobs that arrive early without benefiting any specific process.

SRT equals SPN here because no shorter job arrives while a longer one is executing, i.e., preemption never triggers.

Example 2

Round-Robin quantum: $q = 2$. Total service time = 20. Processes: 3.

Process Set

Process	Arrival	Service
A	0	8
B	1	4
C	6	8

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█	█	█	█	█	█												
B		█							█	█	█	█								
C							█						█	█	█	█	█	█	█	█

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(8+0)/8 = 1.00$
B	$(4+7)/4 = 2.75$
C	$(8+6)/8 = 1.75$
Average	1.83

Round-Robin (RR, $q = 2$)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█			█	█					█	█			█	█				
B		█	█	█			█	█												
C							█		█	█			█	█			█	█	█	█

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(8+8)/8 = 2.00$
B	$(4+3)/4 = 1.75$
C	$(8+6)/8 = 1.75$
Average	1.83

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(8+0)/8 = 1.00$
B	$(4+7)/4 = 2.75$
C	$(8+6)/8 = 1.75$
Average	1.83

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(8+4)/8 = 1.50$
B	$(4+0)/4 = 1.00$
C	$(8+6)/8 = 1.75$
Average	1.42

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(8+0)/8 = 1.00$
B	$(4+7)/4 = 2.75$
C	$(8+6)/8 = 1.75$
Average	1.83

Algorithm Comparison for Example 2

Algorithm	Average Normalized Turnaround Time
FCFS	1.83
RR	1.83
SPN	1.83
SRT	1.42
HRRN	1.83

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.42), while HRRN (avg NTAT = 1.83) is the worst. SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 3

Round-Robin quantum: $q = 4$. Total service time = 15. Processes: 4.

Process Set

Process	Arrival	Service
A	0	4
B	2	6
C	4	2

Process	Arrival	Service
D	7	3

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															
D															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+2)/6 = 1.33$
C	$(2+6)/2 = 4.00$
D	$(3+5)/3 = 2.67$
Average	2.25

Round-Robin (RR, q = 4)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															
D															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+7)/6 = 2.17$
C	$(2+4)/2 = 3.00$
D	$(3+3)/3 = 2.00$
Average	2.04

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															
D															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+4)/6 = 1.67$
C	$(2+0)/2 = 1.00$
D	$(3+5)/3 = 2.67$
Average	1.58

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															
D															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+7)/6 = 2.17$
C	$(2+0)/2 = 1.00$
D	$(3+0)/3 = 1.00$
Average	1.29

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															
C															
D															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+2)/6 = 1.33$
C	$(2+6)/2 = 4.00$
D	$(3+5)/3 = 2.67$
Average	2.25

Algorithm Comparison for Example 3

Algorithm	Average Normalized Turnaround Time
FCFS	2.25
RR	2.04
SPN	1.58
SRT	1.29
HRRN	2.25

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.29), while HRRN (avg NTAT = 2.25) is the worst.

RR ($q = 4$) outperforms FCFS because preemption gives short, late-arriving jobs a chance at the CPU instead of queuing behind long predecessors.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, which is visible in the timeline as short tasks executing soon after arrival.

Example 4

Round-Robin quantum: $q = 1$. Total service time = 20. Processes: 4.

Process Set

Process	Arrival	Service
A	0	6
B	3	2
C	5	8
D	9	4

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█	█	█	█														
B			█				█	█												
C						█			█	█	█	█	█	█	█	█				
D									█								█	█	█	█

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(6+0)/6 = 1.00$
B	$(2+3)/2 = 2.50$
C	$(8+3)/8 = 1.38$
D	$(4+7)/4 = 2.75$
Average	1.91

Round-Robin (RR, $q = 1$)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█		█			█		█										
B			█	█		█														
C						█	█		█			█		█		█		█	█	█
D									█	█		█			█		█			

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(6+4)/6 = 1.67$
B	$(2+1)/2 = 1.50$
C	$(8+7)/8 = 1.88$
D	$(4+4)/4 = 2.00$
Average	1.76

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█	█	█	█														
B			█				█	█												
C						█			█	█	█	█	█	█	█	█				
D									█								█	█	█	█

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(6+0)/6 = 1.00$
B	$(2+3)/2 = 2.50$
C	$(8+3)/8 = 1.38$
D	$(4+7)/4 = 2.75$
Average	1.91

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█			█	█	█												
B			█	█	█															
C						█			█					█	█	█	█	█	█	█
D									█	█	█	█	█							

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(6+2)/6 = 1.33$
B	$(2+0)/2 = 1.00$
C	$(8+7)/8 = 1.88$
D	$(4+0)/4 = 1.00$
Average	1.30

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				
D																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(6+0)/6 = 1.00$
B	$(2+3)/2 = 2.50$
C	$(8+3)/8 = 1.38$
D	$(4+7)/4 = 2.75$
Average	1.91

Algorithm Comparison for Example 4

Algorithm	Average Normalized Turnaround Time
FCFS	1.91
RR	1.76
SPN	1.91
SRT	1.30
HRRN	1.91

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.30), while HRRN (avg NTAT = 1.91) is the worst.

RR ($q = 1$) outperforms FCFS because preemption gives short, late-arriving jobs a chance at the CPU instead of queuing behind long predecessors.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 5

Round-Robin quantum: $q = 2$. Total service time = 25. Processes: 4.

Process Set

Process	Arrival	Service
A	0	9
B	2	5
C	6	7
D	10	4

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█	█	█	█	█	█																
B		█								█	█	█	█	█											
C						█									█	█	█	█	█	█					
D										█											█	█	█	█	

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(9+0)/9 = 1.00$
B	$(5+7)/5 = 2.40$
C	$(7+8)/7 = 2.14$
D	$(4+11)/4 = 3.75$
Average	2.32

Round-Robin (RR, $q = 2$)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█			█	█					█	█						█	█					█	
B		█	█	█			█	█					█												
C						█			█	█					█	█					█	█		█	
D										█				█	█					█	█				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(9+15)/9 = 2.67$
B	$(5+6)/5 = 2.20$
C	$(7+12)/7 = 2.71$
D	$(4+7)/4 = 2.75$
Average	2.58

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█	█	█	█	█	█																
B			█							█	█	█	█	█											
C							█												█	█	█	█	█	█	█
D											█				█	█	█	█							

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(9+0)/9 = 1.00$
B	$(5+7)/5 = 2.40$
C	$(7+12)/7 = 2.71$
D	$(4+4)/4 = 2.00$
Average	2.03

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█						█	█	█	█	█	█	█											
B			█	█	█	█	█																		
C							█												█	█	█	█	█	█	█
D											█				█	█	█	█							

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(9+5)/9 = 1.56$
B	$(5+0)/5 = 1.00$
C	$(7+12)/7 = 2.71$
D	$(4+4)/4 = 2.00$
Average	1.82

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A																									
B																									
C																									
D																									

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(9+0)/9 = 1.00$
B	$(5+7)/5 = 2.40$
C	$(7+8)/7 = 2.14$
D	$(4+11)/4 = 3.75$
Average	2.32

Algorithm Comparison for Example 5

Algorithm	Average Normalized Turnaround Time
FCFS	2.32
RR	2.58
SPN	2.03
SRT	1.82
HRRN	2.32

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.82), while RR (avg NTAT = 2.58) is the worst.

RR (q = 2) performs worse than FCFS here because the quantum induces frequent context switching, lengthening the turnaround time of short jobs that arrive early, without benefiting any specific process.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 6

Round-Robin quantum: $q = 4$. Total service time = 20. Processes: 5.

Process Set

Process	Arrival	Service
A	0	3
B	1	7
C	4	2
D	6	6
E	8	2

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█																	
B		█		█	█	█	█	█	█	█										
C				█							█	█								
D							█						█	█	█	█	█	█		
E									█										█	█

Per-process metrics:

Process	NTAT $((\text{Service} + \text{Waiting}) / \text{Service})$
A	$(3+0)/3 = 1.00$
B	$(7+2)/7 = 1.29$
C	$(2+6)/2 = 4.00$
D	$(6+6)/6 = 2.00$
E	$(2+10)/2 = 6.00$
Average	2.86

Round-Robin (RR, q = 4)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				
D																				
E																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(3+0)/3 = 1.00$
B	$(7+8)/7 = 2.14$
C	$(2+3)/2 = 2.50$
D	$(6+8)/6 = 2.33$
E	$(2+8)/2 = 5.00$
Average	2.60

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A																				
B																				
C																				
D																				
E																				

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(3+0)/3 = 1.00$
B	$(7+2)/7 = 1.29$
C	$(2+6)/2 = 4.00$
D	$(6+8)/6 = 2.33$
E	$(2+4)/2 = 3.00$
Average	2.32

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█																	
B		█		█			█	█			█	█	█	█						
C				█	█	█														
D							█								█	█	█	█	█	█
E									█	█										

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(3+0)/3 = 1.00$
B	$(7+6)/7 = 1.86$
C	$(2+0)/2 = 1.00$
D	$(6+8)/6 = 2.33$
E	$(2+0)/2 = 1.00$
Average	1.44

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	█	█	█																	
B		█		█	█	█	█	█	█	█										
C				█							█	█								
D							█								█	█	█	█	█	█
E									█				█	█						

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(3+0)/3 = 1.00$
B	$(7+2)/7 = 1.29$
C	$(2+6)/2 = 4.00$
D	$(6+8)/6 = 2.33$
E	$(2+4)/2 = 3.00$
Average	2.32

Algorithm Comparison for Example 6

Algorithm	Average Normalized Turnaround Time
FCFS	2.86
RR	2.60
SPN	2.32
SRT	1.44
HRRN	2.32

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.44), while FCFS (avg NTAT = 2.86) is the worst.

RR ($q = 4$) outperforms FCFS because preemption gives short, late-arriving jobs a chance at the CPU instead of queuing behind long predecessors.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 7

Round-Robin quantum: $q = 1$. Total service time = 25. Processes: 5.

Process Set

Process	Arrival	Service
A	0	5
B	2	9
C	3	3
D	7	6
E	11	2

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█																								
B		█																							
C			█																						
D							█																		
E											█														

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(9+3)/9 = 1.33$
C	$(3+11)/3 = 4.67$
D	$(6+10)/6 = 2.67$
E	$(2+12)/2 = 7.00$
Average	3.33

Round-Robin (RR, q = 1)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█																								
B			█																						
C				█																					
D																									
E																									

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+6)/5 = 2.20$
B	$(9+14)/9 = 2.56$
C	$(3+6)/3 = 3.00$
D	$(6+11)/6 = 2.83$
E	$(2+5)/2 = 3.50$
Average	2.82

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█																								
B			█																						
C				█																					
D																									
E																									

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(9+14)/9 = 2.56$
C	$(3+2)/3 = 1.67$
D	$(6+1)/6 = 1.17$
E	$(2+3)/2 = 2.50$
Average	1.78

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█	█																				
B			█														█	█	█	█	█	█	█	█	█
C				█		█	█	█																	
D								█	█	█	█			█	█	█									
E												█	█												

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(9+14)/9 = 2.56$
C	$(3+2)/3 = 1.67$
D	$(6+3)/6 = 1.50$
E	$(2+0)/2 = 1.00$
Average	1.54

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█	█																				
B			█						█	█	█	█	█	█	█	█	█								
C				█		█	█	█																	
D								█												█	█	█	█	█	█
E												█						█	█						

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(5+0)/5 = 1.00$
B	$(9+6)/9 = 1.67$
C	$(3+2)/3 = 1.67$
D	$(6+12)/6 = 3.00$
E	$(2+6)/2 = 4.00$
Average	2.27

Algorithm Comparison for Example 7

Algorithm	Average Normalized Turnaround Time
FCFS	3.33
RR	2.82
SPN	1.78
SRT	1.54
HRRN	2.27

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.54), while FCFS (avg NTAT = 3.33) is the worst. RR ($q = 1$) outperforms FCFS because preemption gives short, late-arriving jobs a chance at the CPU instead of queuing behind long predecessors.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 8

Round-Robin quantum: $q = 2$. Total service time = 30. Processes: 5.

Process Set

Process	Arrival	Service
A	0	10
B	4	4
C	5	8
D	9	5
E	12	3

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A																														
B																														
C																														
D																														
E																														

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(10+0)/10 = 1.00$
B	$(4+6)/4 = 2.50$
C	$(8+9)/8 = 2.12$
D	$(5+13)/5 = 3.60$
E	$(3+15)/3 = 6.00$
Average	3.04

Round-Robin (RR, q = 2)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A																														
B																														
C																														
D																														
E																														

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(10+12)/10 = 2.20$
B	$(4+4)/4 = 2.00$
C	$(8+17)/8 = 3.12$
D	$(5+14)/5 = 3.80$
E	$(3+12)/3 = 5.00$
Average	3.23

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█	█	█	█	█	█	█																				
B				█							█	█	█	█																
C					█																	█	█	█	█	█	█	█	█	█
D									█								█	█	█	█	█									
E												█			█	█	█													

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(10+0)/10 = 1.00$
B	$(4+6)/4 = 2.50$
C	$(8+17)/8 = 3.12$
D	$(5+8)/5 = 2.60$
E	$(3+2)/3 = 1.67$
Average	2.18

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█						█	█	█	█	█																
B				█		█	█	█																						
C					█																	█	█	█	█	█	█	█	█	█
D									█								█	█	█	█	█									
E												█			█	█	█													

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(10+4)/10 = 1.40$
B	$(4+0)/4 = 1.00$
C	$(8+17)/8 = 3.12$
D	$(5+8)/5 = 2.60$
E	$(3+2)/3 = 1.67$
Average	1.96

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A																														
B																														
C																														
D																														
E																														

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(10+0)/10 = 1.00$
B	$(4+6)/4 = 2.50$
C	$(8+9)/8 = 2.12$
D	$(5+16)/5 = 4.20$
E	$(3+10)/3 = 4.33$
Average	2.83

Algorithm Comparison for Example 8

Algorithm	Average Normalized Turnaround Time
FCFS	3.04
RR	3.23
SPN	2.18
SRT	1.96
HRRN	2.83

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.96), while RR (avg NTAT = 3.23) is the worst.

RR (q = 2) performs worse than FCFS here because the quantum induces frequent context switching, lengthening the turnaround time of short jobs that arrive early, without benefiting any specific process.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 9

Round-Robin quantum: $q = 4$. Total service time = 25. Processes: 6.

Process Set

Process	Arrival	Service
A	0	4
B	1	6
C	3	3
D	5	7
E	8	2
F	10	3

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█																					
B		█			█	█	█	█	█	█															
C				█							█	█	█												
D						█								█	█	█	█	█	█	█					
E									█											█	█				
F											█												█	█	█

Per-process metrics:

Process	NTAT $((\text{Service} + \text{Waiting}) / \text{Service})$
A	$(4+0)/4 = 1.00$
B	$(6+3)/6 = 1.50$
C	$(3+7)/3 = 3.33$
D	$(7+8)/7 = 2.14$
E	$(2+12)/2 = 7.00$
F	$(3+12)/3 = 5.00$
Average	3.33

Round-Robin (RR, q = 4)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█																					
B		█			█	█	█	█										█	█						
C			█						█	█	█														
D					█							█	█	█	█								█	█	█
E								█								█	█								
F										█										█	█	█			

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+12)/6 = 3.00$
C	$(3+5)/3 = 2.67$
D	$(7+13)/7 = 2.86$
E	$(2+7)/2 = 4.50$
F	$(3+9)/3 = 4.00$
Average	3.00

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█																					
B		█						█	█	█	█	█	█												
C			█		█	█	█																		
D					█														█	█	█	█	█	█	█
E								█						█	█										
F										█						█	█	█							

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+6)/6 = 2.00$
C	$(3+1)/3 = 1.33$
D	$(7+13)/7 = 2.86$

Process	NTAT ((Service+Waiting)/Service)
E	$(2+5)/2 = 3.50$
F	$(3+5)/3 = 2.67$
Average	2.23

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█																					
B		█						█						█	█	█	█	█							
C			█		█	█	█																		
D						█													█	█	█	█	█	█	█
E									█	█															
F											█	█	█												

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+11)/6 = 2.83$
C	$(3+1)/3 = 1.33$
D	$(7+13)/7 = 2.86$
E	$(2+0)/2 = 1.00$
F	$(3+0)/3 = 1.00$
Average	1.67

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	█	█	█	█																					
B		█			█	█	█	█	█	█															
C			█								█	█	█												
D						█													█	█	█	█	█	█	█
E									█					█	█										
F											█					█	█	█							

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(4+0)/4 = 1.00$
B	$(6+3)/6 = 1.50$
C	$(3+7)/3 = 3.33$
D	$(7+13)/7 = 2.86$
E	$(2+5)/2 = 3.50$
F	$(3+5)/3 = 2.67$
Average	2.48

Algorithm Comparison for Example 9

Algorithm	Average Normalized Turnaround Time
FCFS	3.33
RR	3.00
SPN	2.23
SRT	1.67
HRRN	2.48

Observations

The best algorithm for this workload is SRT (avg NTAT = 1.67), while FCFS (avg NTAT = 3.33) is the worst.

RR ($q = 4$) outperforms FCFS because preemption gives short, late-arriving jobs a chance at the CPU instead of queuing behind long predecessors.

SRT improves on SPN by preempting the running long job whenever a shorter one arrives, as seen in the timeline, with short tasks executing soon after arrival.

Example 10

Round-Robin quantum: $q = 1$. Total service time = 30. Processes: 6.

Process Set

Process	Arrival	Service
A	0	7
B	2	5
C	4	8
D	6	3
E	9	4
F	13	3

First-Come-First-Served (FCFS)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█	█	█	█																							
B		█						█	█	█	█	█																		
C				█									█	█	█	█	█	█	█	█										
D							█														█	█	█							
E									█															█	█	█	█			
F													█															█	█	█

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(7+0)/7 = 1.00$
B	$(5+5)/5 = 2.00$
C	$(8+8)/8 = 2.00$
D	$(3+14)/3 = 5.67$
E	$(4+14)/4 = 4.50$
F	$(3+14)/3 = 5.67$
Average	3.47

Round-Robin (RR, q = 1)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█		█			█				█					█						█								
B			█	█		█		█				█					█													
C					█		█			█					█						█				█			█	█	█
D							█			█			█							█										
E									█				█					█						█			█			
F													█					█				█			█					

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(7+15)/7 = 3.14$
B	$(5+10)/5 = 3.00$
C	$(8+18)/8 = 3.25$
D	$(3+11)/3 = 4.67$
E	$(4+14)/4 = 4.50$
F	$(3+10)/3 = 4.33$
Average	3.82

Shortest Process Next (SPN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█	█	█	█																							
B		█															█	█	█	█	█									
C				█																		█	█	█	█	█	█	█	█	█
D						█	█	█																						
E									█	█	█	█																		
F													█	█	█															

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(7+0)/7 = 1.00$
B	$(5+15)/5 = 4.00$
C	$(8+18)/8 = 3.25$
D	$(3+1)/3 = 1.33$
E	$(4+1)/4 = 1.25$
F	$(3+1)/3 = 1.33$
Average	2.03

Shortest Remaining Time (SRT)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█	█	█	█																							
B		█																█	█	█	█	█								
C				█																		█	█	█	█	█	█	█	█	█
D							█	█	█																					
E									█		█	█	█	█																
F													█		█	█	█													

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(7+0)/7 = 1.00$
B	$(5+15)/5 = 4.00$
C	$(8+18)/8 = 3.25$
D	$(3+1)/3 = 1.33$
E	$(4+1)/4 = 1.25$
F	$(3+1)/3 = 1.33$
Average	2.03

Highest Response Ratio Next (HRRN)

Schedule (each cell represents one time unit; row owner runs in that unit):

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	█	█	█	█	█	█	█																							
B		█						█	█	█	█	█																		
C				█																		█	█	█	█	█	█	█	█	█
D							█						█	█	█															
E									█							█	█	█	█											
F													█							█	█	█								

Per-process metrics:

Process	NTAT ((Service+Waiting)/Service)
A	$(7+0)/7 = 1.00$
B	$(5+5)/5 = 2.00$
C	$(8+18)/8 = 3.25$
D	$(3+6)/3 = 3.00$
E	$(4+6)/4 = 2.50$
F	$(3+6)/3 = 3.00$
Average	2.46

Algorithm Comparison for Example 10

Algorithm	Average Normalized Turnaround Time
FCFS	3.47
RR	3.82
SPN	2.03
SRT	2.03
HRRN	2.46

Observations

The best algorithm for this workload is SPN (avg NTAT = 2.03), while RR (avg NTAT = 3.82) is the worst.

RR ($q = 1$) performs worse than FCFS here because the quantum induces frequent context switching, lengthening the turnaround time of short jobs that arrive early, without benefiting any specific process.

SRT equals SPN here because no shorter job arrives during the execution of a longer one — preemption never triggers.

Critical Analysis and Conclusions

Aggregate Comparison Across All 10 Examples

The table below collects the average normalized turnaround time (NTAT) for every (example, algorithm) pair. The lowest value in each row is highlighted in green; the bottom row shows the unweighted mean of avg NTAT across all 10 examples for each algorithm.

Example	FCFS	RR	SPN	SRT	HRRN
1	1.48	1.74	1.48	1.48	1.48
2	1.83	1.83	1.83	1.42	1.83
3	2.25	2.04	1.58	1.29	2.25
4	1.91	1.76	1.91	1.30	1.91
5	2.32	2.58	2.03	1.82	2.32
6	2.86	2.60	2.32	1.44	2.32
7	3.33	2.82	1.78	1.54	2.27
8	3.04	3.23	2.18	1.96	2.83
9	3.33	3.00	2.23	1.67	2.48
10	3.47	3.82	2.03	2.03	2.46
Mean	2.582	2.542	1.937	1.595	2.215

Key Findings

1. SRT is the strongest performer overall.

SRT achieves the lowest average NTAT in 9 of the 10 examples (it ties with SPN in Example 10) and the lowest mean across all examples. This is consistent with the textbook result that SRT minimizes average turnaround time when service times are known. By preempting whenever a shorter job arrives, SRT prevents short jobs from queuing behind a long job that is already running, i.e., the situation in which FCFS performs worst.

2. FCFS deteriorates sharply as the workload mixes short and long jobs.

FCFS yields acceptable NTAT in Examples 1 and 2 (workloads where the longest job arrives first and there are few short jobs) but has the worst mean across all examples. Examples 7, 9, and 10, each with multiple short jobs arriving after a long one, show FCFS exceeding NTAT = 3.3, dominated by single processes with NTAT values of 5 to 7 (e.g., Example 7 process E with NTAT = 7.0). This is the classic convoy effect.

3. HRRN is the best non-preemptive compromise.

HRRN's mean NTAT is lower than FCFS and RR and close to SPN. Unlike SPN, HRRN has a built-in starvation guard because the waiting term w in $(w + s)/s$ grows linearly with time. In Example 7, HRRN gives an avg NTAT of 2.27, while SPN gives 1.78. SPN is slightly better here on aggregate, but SPN's long-job process B has NTAT = 2.56 (TAT = 23 with service 9); were the workload extended, SPN's bias against B would compound, while HRRN would keep it bounded. HRRN is the policy of choice when fairness matters and service times are known.

4. RR's performance is sensitive to quantum size and workload.

RR with $q = 1$ (Examples 1, 4, 7, 10) tends to outperform FCFS because preemption serves short arrivals quickly, but underperforms SRT because it cannot exploit knowledge of service time. RR with $q = 1$ is also

penalized in Examples 8 and 10, where it slightly exceeds FCFS. The fine-grained rotation distributes the long job's pain across all short jobs, making no process happy. With $q = 4$ (Examples 3, 6, 9), RR behaves more like FCFS because most processes complete within one quantum. Quantum tuning is a real engineering decision; this exercise illustrates why no universal optimum exists.

5. SPN and SRT differ only when shorter jobs arrive mid-execution.

In Examples 1 and 10, SPN equals SRT because no shorter job arrives while a longer one is running — the preemption decision never triggers. In Example 6, however, SRT (1.44) substantially beats SPN (2.32) because the long process B (service 7, arrives at $t = 1$) is preempted multiple times for shorter arrivals C, A, and E.

Recommendations and Trade-offs

Selecting a scheduling policy is rarely a matter of picking the algorithm with the lowest average NTAT on a benchmark. The following considerations frame the choice in practice:

- Interactive systems (responsiveness dominates): RR with a small quantum, tuned so that a typical interactive burst completes within one quantum. The Linux CFS and Windows scheduler descend from this lineage, with refinements in priority and dynamic quantum.
- Batch systems with known job sizes (throughput dominates): SRT or SPN. Used historically in mainframe job scheduling, where service times are estimated from job class declarations.
- Systems requiring fairness with limited starvation: HRRN or its aging-based variants. The aging idea, increasing priority with wait time, appears in nearly every modern OS scheduler.
- Systems with no service-time information: RR, FCFS, or multi-level feedback queues. SPN and SRT are unimplementable without estimates.

Limitations of This Exercise Set

The assumptions made here, single CPU, no I/O, zero context-switching cost, perfect knowledge of service times for SPN and SRT, are not realistic in any production system. Real schedulers must contend with: (i) I/O bursts that change CPU residency patterns, (ii) cache effects that make context switching expensive (often 10–100 μ s of effective overhead), (iii) priority inversion and lock contention, and (iv) the absence of reliable service-time predictions. The values in the master table should therefore be read as upper-bound performance of each algorithm on idealized inputs, not as predictions of relative performance in a real OS. Nonetheless, the qualitative ordering, SRT and SPN best for turnaround, HRRN balanced, FCFS worst on mixed workloads, is robust and matches empirical scheduler studies.

Pedagogical Takeaways

These 10 examples were designed to illustrate, respectively: simple preemption (1), the cost of a long head-of-queue job (2), the FCFS-RR overlap when q matches typical service time (3), RR fragmentation (4), the SPN/SRT advantage for short jobs (5), mixed workloads (6), strong SRT preemption opportunities (7), longer aggregate workload (8), many varied arrivals (9), and dense fine-grained scheduling (10). Working through each by hand and then comparing the average NTAT table builds intuition for why production schedulers use multi-level feedback queues, they approximate SRT's responsiveness for short jobs without requiring service-time prophecy, while bounding the wait of long jobs through aging, the same intuition behind HRRN.